

SELECTED REVISION MATERIAL FROM PREVIOUS COURSES

CONCEPTS FROM SET THEORY

Definition. A pair $(S, \leq)$ is a *partially ordered set* if S is a set and $\leq$ is a partial order on S , that is:

- (i) if $a \leq b$ and $b \leq a$, then $a = b$;
- (ii) we have $a \leq a$ for all $a \in S$;
- (iii) if $a \leq b$ and $b \leq c$, then $a \leq c$.

An example of a partially ordered set is the power set $\mathcal{P}(A)$ of a set A , with subset inclusion $\subseteq$ as a partial order.

Definition. A subset $C \subseteq S$ of a set S is called a *chain* if it is totally ordered, that is, for all $a, b \in C$ either $a \leq b$ or $b \leq a$.

An element $u \in C$ is an *upper bound* for C if $a \leq u$ for all $a \in C$.

An element $m \in S$ is a *maximal element* if $a \in S$ with $m \leq a$ implies $m = a$.

Zorn's Lemma. *If every chain in $(S, \leq)$ has an upper bound, then S has a maximal element.*

Axiom of Choice. *Let $(X_\alpha)_{\alpha \in \Lambda}$ be a set of non-empty sets. Then there exists a choice function*

$$\begin{aligned} f : \Lambda &\rightarrow \bigcup_{\alpha \in \Lambda} X_\alpha \\ \alpha &\mapsto a_\alpha \end{aligned}$$

such that $x_\alpha \in X_\alpha$ for all $\alpha \in \Lambda$.

CONCEPTS FROM GROUP THEORY

Definition. A *group* is a set G together with an operation $*$ on G such that each of the following holds.

- (i) Closure: $a * b \in G$ for all $a, b \in G$.
- (ii) Associativity: $a * (b * c) = (a * b) * c$ for all $a, b, c \in G$.
- (iii) Existence of an identity element: there is an identity element $e \in G$ such that $a * e = e * a = a$ for each $a \in G$.
- (iv) Existence of inverse elements: for each $a \in G$ there is an element $b \in G$ such that $a * b = b * a = e$.

The above definition is often more concisely written as: A *group* is a set G together with a binary operation on G such that each of the following holds.

- (i) Closure: $ab \in G$ for all $a, b \in G$.
- (ii) Associativity: $a(bc) = (ab)c$ for all $a, b, c \in G$.
- (iii) Existence of an identity element: There is an identity element $e \in G$ such that $ae = ea = a$ for each $a \in G$.
- (iv) Existence of inverse elements: For each $a \in G$ there is an element $b \in G$ such that $ab = ba = e$.

Definition. If G and H are groups, then we may define a new group $G \times H$, called the *direct product* of G and H , to be the set $\{(g, h) \mid g \in G, h \in H\}$ with operation

$$(g_1, h_1) \cdot (g_2, h_2) = (g_1 g_2, h_1 h_2).$$

Theorem (Quick Subgroup Theorem). *Let G be a group with operation $*$, and let H be a subset of G . Then H is a subgroup of G if and only if*

- (a) H is non-empty;
- (b) if $a \in H$ and $b \in H$, then $a * b \in H$; and
- (c) if $a \in H$, then $a^{-1} \in H$.

Theorem (First Isomorphism Theorem). *Let G and H be groups, and let $f : G \rightarrow H$ be a homomorphism. Then $\ker f \trianglelefteq G$ and $\operatorname{im} f \cong G / \ker f$.*

Theorem (Second Isomorphism Theorem). *Let G be a group, let H be a subgroup, and let N be a normal subgroup of G . Then $HN = \{hn : h \in H, n \in N\}$ is a subgroup of G , and $H \cap N \trianglelefteq H$, and*

$$HN/N \cong H/H \cap N.$$

Theorem (Third Isomorphism Theorem). *If $K \trianglelefteq G$ and $K \leq L \trianglelefteq G$, then*

$$G/K/L/K \cong G/L.$$

Theorem (Correspondence Theorem). *Let G be a group, and let N a normal subgroup of G . Then every subgroup of the quotient group G/N is of the form H/N , where $N \leq H \leq G$. Conversely, if $N \leq H \leq G$ then $H/N \leq G/N$.*

Theorem (Lagrange). *Let H be a subgroup of a finite group G . Then the order of H is a divisor of the order of G .*

Theorem (Fundamental Theorem of Finite Abelian Groups). *Let A be a finitely generated abelian group. Then*

$$(1) \quad A \cong \mathbb{Z}^s \oplus \mathbb{Z}/a_1\mathbb{Z} \oplus \cdots \oplus \mathbb{Z}/a_r\mathbb{Z},$$

where s is a non-negative integer and the a_i 's are non-zero non-units in $\mathbb{Z}$, such that

$$(2) \quad a_1 \mid a_2 \mid \cdots \mid a_r.$$

Further, the decomposition (1) of A subject to the condition (2) is unique. (Here $\mathbb{Z}^0$ is interpreted as the trivial group (0) .)

A presentation of a group G comprises a set X of generators and a set R of relations among those generators. We then say G has presentation

$$\langle X \mid R \rangle$$

and we write $G = \langle X \mid R \rangle$. [To formally and precisely define group presentations, one needs the concept of a *free group* $F(X)$ on a set X , that is, the group generated by the set X , where the only relations are of the form $x \cdot x^{-1} = 1$, for all $x \in X$. The group G is then said to have the presentation $\langle X \mid R \rangle$ if G is isomorphic to the quotient of the free group

on X by the normal subgroup generated by the relations R . If there are no relations, i.e. if $R = \emptyset$, then we simply write $\langle X \rangle$ instead of omit the notation $\langle X \mid \emptyset \rangle$. So the free group $F(X)$ on X has presentation $\langle X \rangle$.]

For example, the cyclic group of order 2 has the presentation $\langle x \mid x^2 = 1 \rangle$ and the integers $\mathbb{Z}$ has the presentation $\langle x \rangle$. Additionally $\mathbb{Z} \times \mathbb{Z} = \langle x, y \mid [x, y] = 1 \rangle$, where $[x, y] = x^{-1}y^{-1}xy$ is called the *commutator* of x and y .

Recall the dihedral group D_{2m} of order $2m$:

$$D_{2m} = \langle x, y \mid x^m = 1 = y^2, yxy = x^{-1} \rangle$$

We can view D_{2m} as a subgroup of $\text{GL}_2(\mathbb{R})$ by taking X to be the counterclockwise rotation by $\frac{2\pi}{m}$ and Y to be the reflection in the x -axis.

CONCEPTS FROM RING THEORY

Definition. A *ring* is a set R together with two operations on R , usually written as *addition* $a + b$ and *multiplication* $a \cdot b$, such that

- (i) R with addition is a group;
- (ii) R is closed with respect to multiplication;
- (iii) multiplication is associative, i.e. $a(bc) = (ab)c$ for every $a, b, c \in R$;
- (iv) multiplication is distributive over addition, i.e.

$$a(b + c) = ab + ac \quad \text{and} \quad (a + b)c = ac + bc$$

for all $a, b, c \in R$. The above are called the *distributive laws*.

The group formed by R with respect to addition is called the *additive group* of R , and the identity element 0 is called the *zero* of the ring R .

Theorem (Quick Subring Theorem). *A subset S of a ring R is a subring of R if and only if*

- (a) S is non-empty;
- (b) S is closed under both the addition and the multiplication of R ; and
- (c) S contains the negative of each of its elements.

Definition. A subring I of a ring R is a *left ideal* (respectively *right ideal*) of R if $ra \in I$ (respectively $ar \in I$) for all $a \in I$ and all $r \in R$.

Definition. A subring I of a ring R is an *ideal* of R if $ar \in I$ and $ra \in I$ for all $a \in I$ and all $r \in R$. Notation: $I \triangleleft R$

Definition. An element $a \neq 0$ in a commutative ring R is called a *zero divisor* in R if there exists an element $b \neq 0$ in R such that $ab = 0$.

Definition. A commutative ring with unity $e \neq 0$ and no zero divisors is called an *integral domain*.

Definition. Let R be a commutative ring with unity e , and let $a \in R$. Then (a) will denote the set of all multiples of a by elements of R :

$$(a) = \{ra \mid r \in R\}$$

An ideal of the form (a) is called a *principal ideal*.

Definition. An integral domain in which every ideal is a principal ideal is called a *principal ideal domain* (PID).

Definition. Let R be a commutative ring. An ideal $P \neq R$ is *prime* if whenever $a, b \in R$ and $ab \in P$, then $a \in P$ or $b \in P$.

Consider $R = \mathbb{Z}$ and let $P = 2\mathbb{Z} = (2)$. Then if $ab \in 2\mathbb{Z}$, i.e. if ab is even then at least one of a or b has to be even. So P is a prime ideal.

Definition. Let R be a commutative ring. An ideal $M \subsetneq R$ is a *maximal* ideal of R if and only if R has no ideal I such that $M \subsetneq I \subsetneq R$.

Likewise if $R = \mathbb{Z}$ then $M = 2\mathbb{Z}$ is a maximal ideal, since if there is an ideal I such that $M \subsetneq I \subseteq R$, then I contains an odd integer, say $2k + 1$ for some $k \in \mathbb{Z}$. Then $2\mathbb{Z} \subseteq I$ and $2\mathbb{Z} + (2k + 1) \subseteq I$, but

$$\begin{aligned} 2\mathbb{Z} + (2k + 1) &= \{2m + (2k + 1) \mid m \in \mathbb{Z}\} \\ &= \{2(m + k) + 1 \mid m \in \mathbb{Z}\} \\ &= \{2n + 1 \mid n \in \mathbb{Z}\}, \end{aligned}$$

which is equal to the set of all odds integers. So I contains all even and all odd integers, and is hence equal to R . Therefore M is maximal.

Theorem. Let R be a commutative ring with unity. Then each maximal ideal of R is a prime ideal.

Proof. Suppose that M is a maximal ideal in R that is not a prime ideal. Then there exist $a, b \in R$ such that $a \notin M$, $b \notin M$ but $ab \in M$.

Let S be the smallest ideal of R containing M and a , and let T be the smallest ideal of R containing M and b . As $a \notin M$ we have $S = R$. Similarly $T = R$. Hence $ST = RR = R$. However $S = \{m_1 + r_1a \mid m_1 \in M, r_1 \in R\}$ whilst $T = \{m_2 + r_2b \mid m_2 \in M, r_2 \in R\}$. Recall that ST is defined to be the set of sums of elements of the form st , where $s \in S$ and $t \in T$. Hence, each element in ST is a sum of elements of the form $(m_1 + r_1a)(m_2 + r_2b) = m_1m_2 + m_1r_2b + r_1am_2 + abr_1r_2$. Each term in this sum is in M , therefore $(m_1 + r_1a)(m_2 + r_2b) \in M$. Hence $ST \subseteq M \subsetneq R$, which gives us our required contradiction. $\square$

Lemma. Let R be a commutative ring and I an ideal in R . Then R/I is an integral domain if and only if I is a prime ideal in R .

Proof. Suppose that R/I is an integral domain and let $ab \in I$. Then $I = I + ab = (I + a)(I + b)$ and so either $I + a = I$ or $I + b = I$. So $a \in I$ or $b \in I$.

Suppose that I is prime and suppose that $(I + a)(I + b) = I$. Then $I + ab = I$, so $ab \in I$. Primeness implies that $a \in I$ or $b \in I$. In other words, $I + a = I$ or $I + b = I$, which gives us that R/I is an integral domain. $\square$

Let $R = \mathbb{Z}_6[x]$ and consider the ideal $I = (x)$. As $\mathbb{Z}_6 \cong \mathbb{Z}_6[x]/(x)$ is an integral domain, the principal ideal (x) must be prime, by the above lemma.

The following will be stated without proof.

Lemma. Let R be a commutative ring and I an ideal in R . Then R/I is a field if and only if I is maximal in R .

Examples: (i) Let $R = \mathbb{Z}$ and $I = 2\mathbb{Z}$. Here $\mathbb{Z}/2\mathbb{Z}$ is a field and $2\mathbb{Z}$ is maximal (and prime by the above theorem).

(ii) Let $R = \mathbb{Z}[x]$ and $I = (x, p)$ for a prime p . Then $\mathbb{Z}[x]/(x, p) \cong \mathbb{Z}_p$ is a field and (x, p) is a maximal (and prime) ideal in $\mathbb{Z}[x]$.

(iii) Let $R = \mathbb{C}[x]$ and $I = (x)$. Here $\mathbb{C}[x]/(x) \cong \mathbb{C}$ is a field and therefore (x) is maximal (and prime) in $\mathbb{C}[x]$.

Definition. Let D be an integral domain and let $a, b \in D$ with $b \neq 0$. We say that a is *divisible* by b if $a = bc$ for some $c \in D$. If a is divisible by b , we say that b *divides* a , or that b is a factor of a . Notation: $b \mid a$.

Definition. Let D be an integral domain. An element of D is called a *unit* if it divides the unity e of D .

The units of $\mathbb{Z}$ are 1 and -1 .

Let F be a field and consider $F[x]$. The units of $F[x]$ are the polynomials of degree zero (in other words, the non-zero constant polynomials).

Definition. Let D be an integral domain and let $a, b \in D$. The elements a and b are called *associates* if $a \mid b$ and $b \mid a$.

Lemma. Let D be an integral domain with $a, b \in D$. Then a and b are associates if and only if $a = bu$ for some unit $u \in D$.

Elements $a, b \in \mathbb{Z}$ are associates if and only if $a = \pm b$.

Let F be a field and consider $F[x]$. The polynomials $f(x), g(x) \in F[x]$ are associates if and only if $g(x) = c \cdot f(x)$ for some non-zero $c \in F$.

Definition. Let D be an integral domain. An element in an integral domain is said to be *irreducible* if it is not a unit of D and its only divisors in D are its associates and the units of D .

The irreducible elements of $\mathbb{Z}$ are the primes and their negatives.

Let F be a field. The irreducible elements of $F[x]$ are the irreducible polynomials of $F[x]$.

Definition. Let R be an integral domain. An element $p \in R$ is a *prime* (or *prime element*) if $p \neq 0$, p is a non-unit and if $p \mid ab$ then either $p \mid a$ or $p \mid b$.

In an integral domain, every prime is irreducible but the converse is not true in general. For example, in the integral domain $\mathbb{Z}[\sqrt{-5}]$, the number 3 is irreducible but not prime (as 3 divides $9 = (2 + \sqrt{-5})(2 - \sqrt{-5})$ but 3 does not divide any factor on the right).

Definition. A *division ring* is a ring where every non-zero element has a multiplicative inverse.

Definition. A commutative ring in which the set of non-zero elements form a group with respect to multiplication is called a *field*.

The ring $\text{GL}_2(\mathbb{R})$ is an example of a division ring that is not a field.

We write R^{op} for the *opposite ring* of a ring R , that is, the ring with the same underlying set but with multiplication defined as $r \cdot s = sr$, for $r, s \in R$, where the right-hand side of the equation is the given multiplication in the original ring R .

Lemma. *Let R be a ring. Then $(M_n(R))^{\text{op}} \cong M_n(R^{\text{op}})$.*

Proof. Define a mapping $f : (M_n(R))^{\text{op}} \rightarrow M_n(R^{\text{op}})$ by $f(A) = A^T$, the transpose of A . Recall that, as sets (and also as additive groups) we have $R = R^{\text{op}}$ and $M_n(R) = (M_n(R))^{\text{op}}$. Then by the definition of the transpose of a matrix,

$$(A + B)^T = A^T + B^T$$

so $f(A+B) = f(A)+f(B)$. We now show that $f(A \circ B) = f(A)f(B)$, where the multiplication of matrices $f(A)$ and $f(B)$ is in the ring $M_n(R^{\text{op}})$. Write $A = (a_{ij})$, $B = (b_{ij})$, $f(A) = A^T = (a'_{ij})$ and $f(B) = B^T = (b'_{ij})$. Then $a'_{ij} = a_{ji}$ and $b'_{ij} = b_{ji}$ for all $1 \leq i, j \leq n$. Now

$$f(A \circ B) = f(BA) = (BA)^T.$$

The (i, j) -entry of $(BA)^T$ is the (j, i) -entry of BA , which is

$$\begin{aligned} \sum_{k=1}^n b_{jk} a_{ki} &= \sum_{k=1}^n a_{ki} \circ b_{jk} = \sum_{k=1}^n a'_{ik} \circ b'_{kj} \\ &= (i, j)\text{-entry of } A^T B^T \in M_n(R^{\text{op}}). \end{aligned}$$

Hence $(BA)^T = A^T B^T$. This proves that $f(A \circ B) = f(A)f(B)$. The mapping f is clearly 1-1 and onto. Hence the result. $\square$

SOME NOTATIONAL ASPECTS

Recall that $\mathbb{N}$ denotes the set of positive integers $\{1, 2, 3, \dots\}$.

Recall that the Kronecker delta δ_{ij} is defined as

$$\delta_{ij} = \begin{cases} 1 & \text{if } i = j, \\ 0 & \text{if } i \neq j. \end{cases}$$

For G a group, we denote by $H < G$ or $H \leq G$ when H is a proper subgroup of G . Similarly for proper subrings and proper subfields. Also for a proper subset S of a set X , we likewise write $S \subset X$ or $S \subsetneq X$.

If G is a group and $H \leq G$ a subgroup, we write $|G : H|$ or $[G : H]$ to denote the index of H in G .

Recall the symmetric group S_n (also denoted $\text{Sym}(n)$) on n objects. Let $X = \{1, 2, \dots, n\}$. Then S_n is the group whose elements are all bijective functions (equivalently, permutations) from X to X and whose group operation is that of function composition (or permutation composition).

A *transposition* is a permutation which exchanges two elements and keeps all others fixed. Every permutation can be written as a product of transpositions.

Recall that a permutation is called *even* if it can be expressed as a product of even number of transpositions. The alternating group A_n (also denoted $\text{Alt}(n)$) is the group of all even permutations of $X = \{1, 2, \dots, n\}$.

Recall that the dihedral group D_8 of order 8 is the group of symmetries of the square.